

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

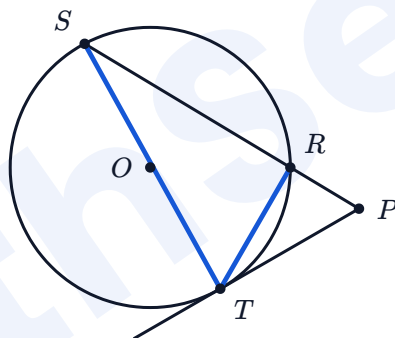
Chapter 18: Tangents and Intersecting Chords

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Tangent and secant from an external point

1. PT is a tangent to a circle at T , and a line through P cuts the circle at R and S , with R nearer to P . (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT . **ICSE 2025, 3 marks**



SOLUTION

(a) In $\triangle PTR$ and $\triangle PST$: $\angle P$ is common, and $\angle PTR = \angle PST$ (angle between tangent and chord equals the angle in the alternate segment). So $\triangle PTR \sim \triangle PST$ (AA).

(b) Corresponding sides are proportional: $\frac{PT}{PS} = \frac{PR}{PT}$, so $PT^2 = PR \times PS$.

(c) $PT^2 = 4 \times 16 = 64$

Answer: $PT = 8$ cm

2. A circle has centre O and radius 9 cm. The diameter AB is produced to a point P , and PQ is a tangent from P . If $PA = 24$ cm, find PQ . **ICSE 2024, 3 marks**

SOLUTION

$AB = 18$ cm, so $PB = PA - AB = 24 - 18 = 6$ cm.

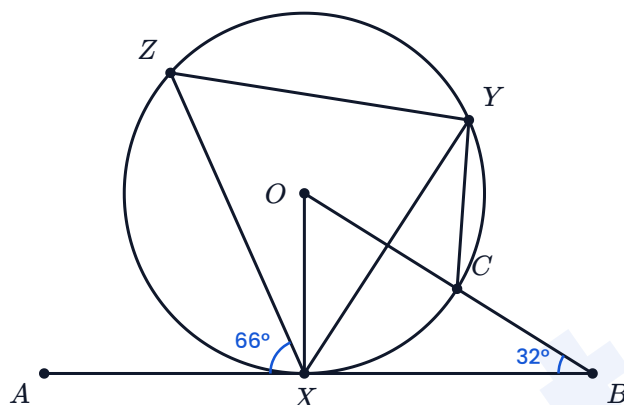
$PQ^2 = PA \times PB = 24 \times 6 = 144$

Answer: $PQ = 12$ cm

(Check: $OP = 15$ cm and $OQ = 9$ cm, so $PQ = \sqrt{15^2 - 9^2} = 12$ cm.)

The alternate segment theorem

3. X, Y, Z and C lie on a circle with centre O . AB is the tangent at X , OB cuts the circle at C , and $ZY = XY$. Given $\angle OBX = 32^\circ$ and $\angle AXZ = 66^\circ$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$.
ICSE 2025, 4 marks

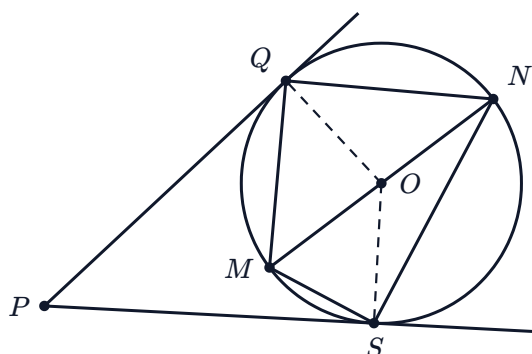


SOLUTION

- (a) $OX \perp AB$ (radius and tangent). In $\triangle OXB$: $\angle BOX = 90^\circ - 32^\circ = 58^\circ$.
 (b) The arc CX subtends $\angle COX = 58^\circ$ at the centre and $\angle CYX$ on the circle: $\angle CYX = \frac{58^\circ}{2} = 29^\circ$.
 (c) By the alternate segment theorem, $\angle ZYX = \angle AXZ = 66^\circ$.
 (d) $ZY = XY$, so $\angle YXZ = \frac{180^\circ - 66^\circ}{2} = 57^\circ$. Also $\angle OXZ = 90^\circ - 66^\circ = 24^\circ$. So $\angle OXY = 57^\circ - 24^\circ = 33^\circ$.

Answer: (a) 58° (b) 29° (c) 66° (d) 33°

4. PR and PT are tangents from P to a circle with centre O , touching it at Q and S . MN is a diameter, with M on the arc QS nearer to P . Given $\angle PQM = 42^\circ$ and $\angle PSM = 25^\circ$, find (a) $\angle OQM$ (b) $\angle QNS$ (c) $\angle QOS$ (d) $\angle QMS$. **ICSE 2024, 4 marks**



SOLUTION

(a) $OQ \perp PR$ (radius and tangent), so $\angle OQM = 90^\circ - 42^\circ = 48^\circ$.

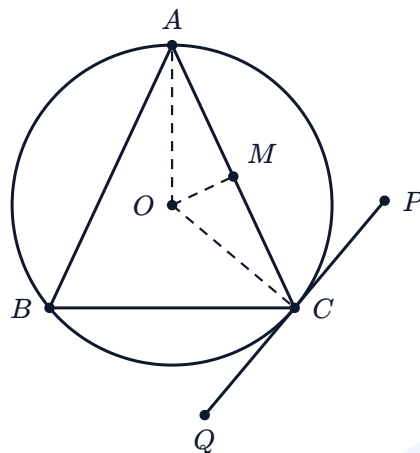
(b) By the alternate segment theorem, $\angle QNM = \angle PQM = 42^\circ$ and $\angle SNM = \angle PSM = 25^\circ$.
So $\angle QNS = 42^\circ + 25^\circ = 67^\circ$.

(c) Angle at the centre is twice the angle at the circumference: $\angle QOS = 2 \times 67^\circ = 134^\circ$.

(d) $QNSM$ is a cyclic quadrilateral: $\angle QMS = 180^\circ - 67^\circ = 113^\circ$.

Answer: (a) 48° (b) 67° (c) 134° (d) 113°

5. An isosceles triangle ABC with $AB = AC$ is inscribed in a circle with centre O . PQ is the tangent at C , with Q on the side of BC away from A . OM is perpendicular to the chord AC and $\angle COM = 65^\circ$. Find (a) $\angle ABC$ (b) $\angle BAC$ (c) $\angle BCQ$. **ICSE 2024, 3 marks**



SOLUTION

(a) The perpendicular from the centre bisects the chord and the angle at the centre:

$$\angle AOC = 2 \times 65^\circ = 130^\circ. \text{ So } \angle ABC = \frac{130^\circ}{2} = 65^\circ.$$

(b) $AB = AC$, so $\angle ACB = \angle ABC = 65^\circ$ and $\angle BAC = 180^\circ - 130^\circ = 50^\circ$.

(c) By the alternate segment theorem, $\angle BCQ = \angle BAC = 50^\circ$.

Answer: (a) 65° (b) 50° (c) 50°

3 Tangent properties

6. PS and PT are tangents to a circle from P , with $\angle SPT = 80^\circ$. Q is a point on the circle with $SQ \parallel PT$. Find $\angle QST$. (a) 140° (b) 90° (c) 80° (d) 50° **ICSE 2024, 1 mark**

SOLUTION

$$PS = PT, \text{ so } \angle PTS = \angle PST = \frac{180^\circ - 80^\circ}{2} = 50^\circ.$$

$SQ \parallel PT$ and ST is a transversal, so $\angle QST = \angle PTS = 50^\circ$ (alternate angles).

Answer: (d)

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